

Chapter 2

1. (a) E (b) 2 (c) 17 (d) A6 (e) 3F0 (f) 982
2. (a) 35 (b) 146 (c) 26 (d) 141 (e) 243 (f) 235 (g) 1474 (h) 1792
3. (a) 10110 (b) 1101111 (c) 1000110011001
4. (a) 1100 (b) 00011 (c) 10000011110

Chapter 3

5. The output waveform X is HIGH during intervals 2, 6, 10, and 14. X is LOW at all other times.
6. The output waveform X is HIGH during intervals 4 and 12. X is LOW at all other times.
7. The output waveform X is LOW during interval 1. X is HIGH from interval 2 through the remainder of the diagram.
8. When input B is LOW (intervals 5-7, 12-14), output X is forced to a constant HIGH. When input B is HIGH (intervals 1-4, 8-11, 15-16), output X is the exact inverted waveform of input A.
9. The output waveform X is LOW during intervals 3, 5, 12, and 14. X is HIGH at all other times.
10. The output waveform X is HIGH during intervals 1, 7, and 13. X is LOW at all other times.
11. From the NAND perspective, both inputs are HIGH only in intervals 1 and 7, which means a LOW output. From the negative OR perspective, at least one input is LOW during all intervals except 1 and 7, which is a HIGH output. Both confirm that output X is LOW during intervals 1 and 7, and HIGH at other times.
12. When input B is LOW output X matches A. When input B is HIGH X is the inverted waveform of A.
13. B is LOW output so X is the inverted waveform of A. When B is HIGH output X matches waveform A.